Model Evaluation Report

Fake job postings. Binary classification of the column scam.

Machine learning goal

The goal is classification, not a numeric forecast. The column to predict is scam: 1 when the trust and safety team confirmed the posting as a scam, and 0 when they confirmed it as legitimate. The file holds 400 reviewed postings. 109 are scams (27.3%) and 291 are legitimate.

A rule that calls every posting legitimate is already right on 291 of the 400 rows, which is 72.8% accuracy, and it catches no scams. Accuracy is therefore reported together with precision, recall, and ROC-AUC, all measured on the scam class.

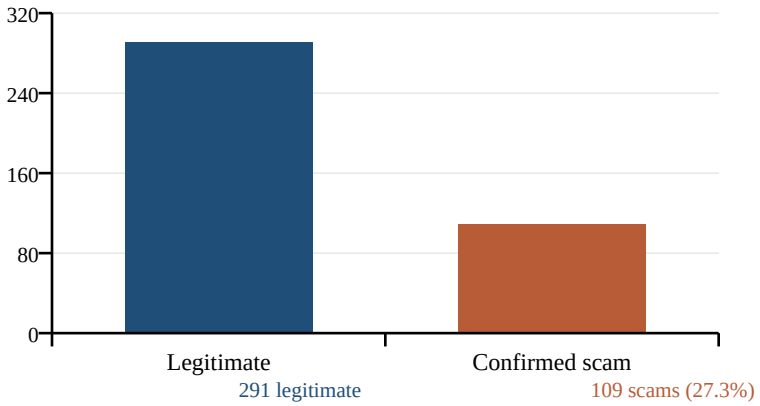

Figure 1. Confirmed labels in the 400 reviewed postings.

Features used

Nine columns were kept from the cleaning stage. A column stayed only when the absolute correlation with scam was at least 0.2. Category, vague-word count, exclamation marks, employer response delay, the remote flag, and a free-email contact address were removed on that rule.

Two cleaning steps are applied before the classifier sees a row. First, 124 postings leave both salary cells empty. Those cells are filled with the median of the rows used to fit the model. On the full file those medians are 4,766.50 and 6,650.50. In the cross-validation below, each fold uses the median of its own training rows only. Second, the nine columns are standardized: the training mean is subtracted and the result is divided by the training standard deviation. No extra column was added for the empty salary.

Feature	Correlation with scam	Coefficient in the saved model
Skills listed	-0.675	-2.107
Experience required (years)	-0.624	-1.887
Minimum salary	-0.332	-1.567
Company size	-0.342	-1.533
Company age (months)	-0.209	-0.701
Hour posted	-0.466	-0.445
Asks for a fee or payment details	+0.220	+0.181
Maximum salary	+0.394	+1.319
Capital-letter ratio	+0.724	+1.522

Salary correlations skip the 124 rows with no published pay, because a correlation cannot use an empty cell. The coefficients come from the model that was fit on all 400 rows and saved. They are on the standardized scale. A negative coefficient pushes the verdict toward legitimate. A positive coefficient pushes it toward scam.

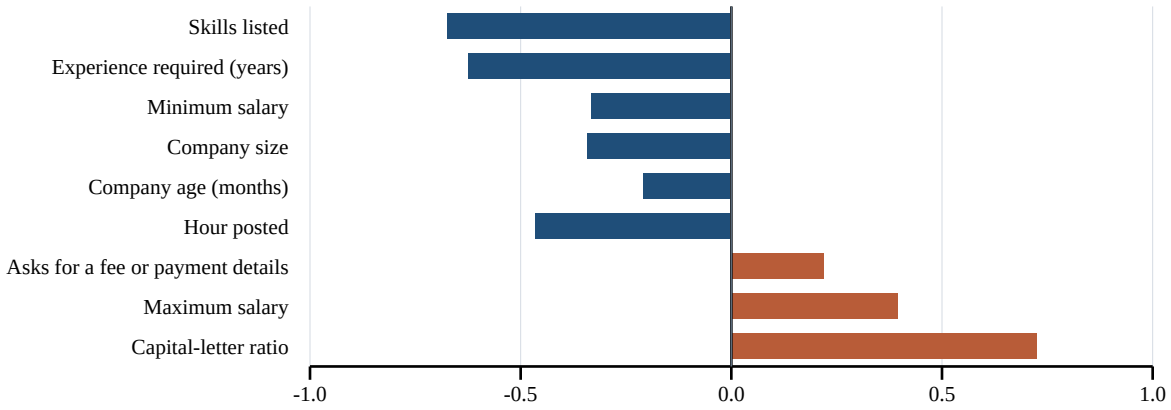

Figure 2. Correlation with scam. Navy bars point toward legitimate postings. Rust bars point toward scams. The 0.2 cutoff is why these nine columns remain.

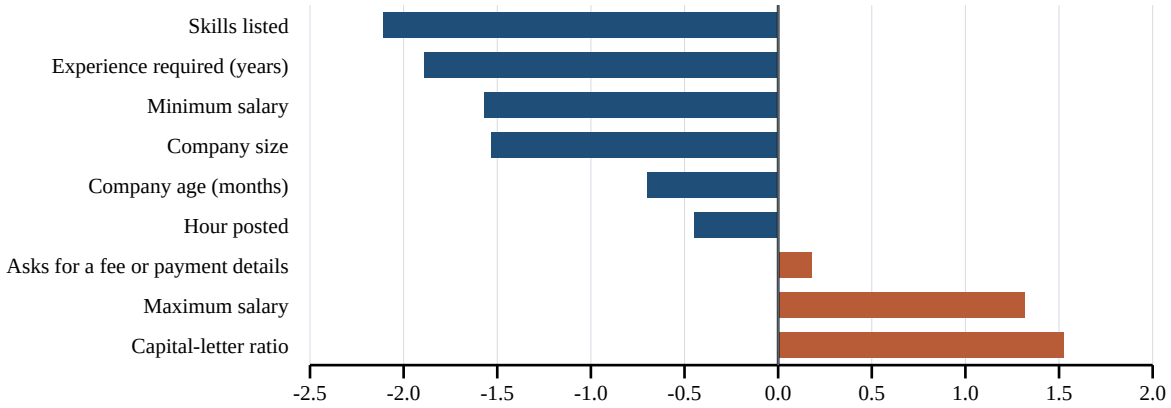

Figure 3. Coefficients in the saved logistic regression, after standardization. The largest pushes are skills listed and years of experience, both toward legitimate.

Model parameters

The verdict is a logistic regression (scikit-learn LogisticRegression). The solver is lbfgs, with L2 regularization at C = 1.0 and a maximum of 1,000 iterations. class_weight is balanced, so the 109 scams count as heavily in the fit as the 291 legitimate postings. The predicted class is scam when the scam probability is at least 0.5.

The same standardized rows are also stored in a nearest-neighbor index with three neighbors and Euclidean distance. That index does not choose the verdict. The Streamlit app uses it only to show the three closest reviewed postings. The metrics in this report are for the logistic regression verdict.

The check described next is done first. The classifier, the scaler, the salary medians, the neighbor index, the 400 reviewed rows, and these scores are then written to model.joblib. The app loads that file. It does not train again.

Evaluation

The check is stratified 5-fold cross-validation, shuffled with random_state 42. Each fold holds out 80 postings, and the scam rate stays near 27% in every fold. Every posting is scored by a model that was not trained on it. Three setups share the same folds.

The baseline calls every posting legitimate. Its scam probability is the training scam rate, a constant, so its ROC-AUC is 0.5. The full model is the same logistic regression trained on every original column, with category turned into indicator columns and the same salary fill, scaling, and class weighting. The saved model is that regression trained on the nine features only.

Setup	Accuracy	Precision	Recall	ROC-AUC
Always legitimate	0.728	0.000	0.000	0.500
Logistic regression, all columns	0.988	0.982	0.973	0.998
Logistic regression, nine features (saved)	0.988	0.982	0.973	0.999

Each number is the mean of the five folds. Precision and recall use the scam class and the 0.5 cutoff. ROC-AUC uses the scam probability.

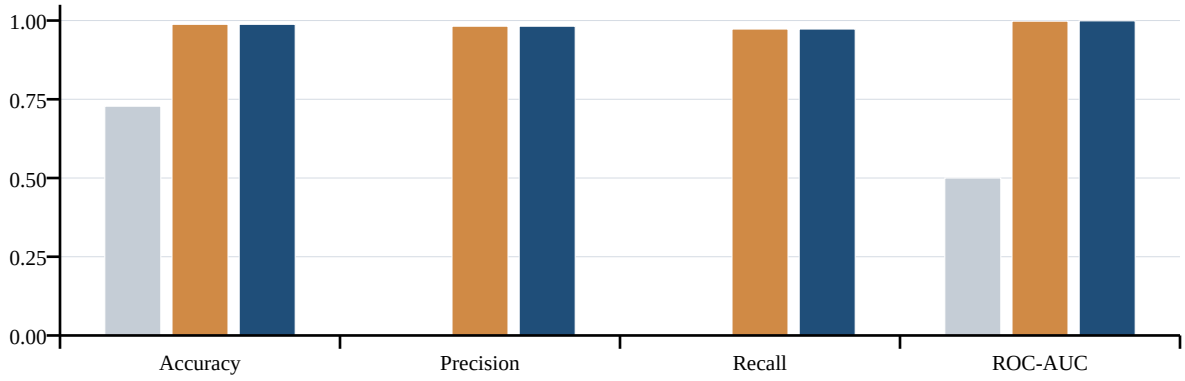

Figure 4. Grey is the always-legitimate rule, orange is every original column, and navy is the saved nine-feature model. The two model bars match at three decimal places. The baseline has no precision and no recall on the scam class.

Interpretation

On the 400 held-out predictions the saved model is correct 395 times (98.8%). It catches 106 of the 109 scams and misses 3. It marks 2 legitimate postings as scams and leaves the other 289 legitimate postings alone. That is a recall of 97.3% and a precision of 98.2%. The 80% accuracy target is met, and the recall of 97.3% shows the scams are being caught. The always-legitimate rule reaches 72.8% accuracy with a recall of zero.

	Called legitimate	Called scam
Actually legitimate	289	2
Actually scam	3	106

Figure 5. Held-out predictions from the five folds, added together. The green cells are the 395 correct calls. The two rust cells are the 2 false alarms and the 3 missed scams.

Training on every original column does not improve on the nine-feature model. Accuracy, precision, and recall are the same at three decimal places. ROC-AUC is 0.999 on the nine features and 0.998 on the full set. The columns removed in cleaning do not add signal once these nine are present. The nine-feature model is the one saved, because a new posting only has to supply those fields.

In the saved model, more listed skills, more years of experience, a higher published minimum salary, and a larger company push the verdict toward legitimate. A higher share of capital letters, and a higher published maximum salary, push it toward scam. Read together, the two salary coefficients say that a low floor and a high ceiling are the pay pattern associated with scams in this file. The fee flag stays positive, but its coefficient is small (+0.181) once the other features are in the model, so it is a weak extra sign rather than the main one.

The classes in this file separate very cleanly. The holdout is still the right check: three confirmed scams are missed, and two legitimate postings would be blocked. Those five cases are the errors of this model on this file. Rebuilding the saved file is python train.py.